

Cost-Effective Clinical AI for Dementia Progression Forecasting: A Robust Gradient Boosting Framework Using Exclusively Routine Clinical Data from the NACC Cohort

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Abstract

Background and Objective: Early identification of patients progressing from Mild Cognitive Impairment (MCI) to Alzheimer’s Disease (AD) is critical for therapeutic intervention. Current state-of-the-art predictive models predominantly rely on expensive neuroimaging (MRI/PET) or invasive cerebrospinal fluid (CSF) biomarkers, limiting accessibility in primary care and resource-limited settings. This study develops a high-performance, cost-effective, and ethically robust machine learning framework to forecast dementia progression using *exclusively* routine clinical data.

Methods: We used longitudinal data from the National Alzheimer’s Coordinating Center (NACC) Uniform Data Set, comprising 150,904 clinical visits from 37,633 patients across 40+ Alzheimer’s Disease Research Centers. We framed this as a “Next-Visit Forecasting” problem, predicting the CDR® Dementia Staging Instrument score (CDR) at time $t + 1$ from data at time t . We developed a robust XGBoost framework for clinical tabular data, evaluating BorderlineSMOTE and a weighted ensemble as ablations, with patient-level GroupShuffleSplit validation to prevent data leakage and audits for demographic fairness and error severity.

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Results: On an independent test set of 29,939 visits from 7,527 patients, the base XGBoost model achieved **87.82%** accuracy, **85.14%** macro F1, and **87.95%** weighted F1. Class-specific F1-scores were 0.93 (Normal), 0.83 (MCI), 0.77 (Mild), and 0.88 (Severe). Of all misclassifications, 94.9% were one stage off, with only 0.2% three stages off. Performance was broadly consistent across sex, age, and race. SHAP analysis identified diagnosis trajectory, current diagnostic state, and age-diagnosis interaction features as the strongest predictors. Adding BorderlineSMOTE or ensembling reduced aggregate performance relative to the base model.

Conclusions: Robust dementia risk stratification is achievable without expensive neuroimaging, using only routine clinical assessments. The framework offers a scalable, transparent, and demographically fair tool for clinical decision support in community healthcare and resource-constrained settings, supporting a “Frugal AI” paradigm that democratizes early Alzheimer’s detection.

Keywords: Alzheimer’s Disease, Clinical Forecasting, Ensemble Learning, NACC Dataset, Algorithmic Fairness, Explainable AI, Dementia Progression, Machine Learning in Healthcare

Highlights

- Routine clinical data forecast next-visit dementia status.
- Patient-level splitting prevents longitudinal data leakage.
- Base XGBoost achieves 87.82% accuracy and 85.14% macro F1.
- A roadmap links model outputs to primary-care EHR workflows.

Summary Table

- A clinical-only XGBoost model predicts four next-visit dementia stages.
- Evaluation used 29,939 visits from 7,527 previously separated patients.
- The study defines prospective validation and EHR translation steps.

1. Introduction

Alzheimer’s disease (AD) is a progressive neurodegenerative disorder and the leading cause of dementia, affecting over 55 million people worldwide [1]. As the global population ages, the prevalence of AD is projected to triple by 2050, imposing an unsustainable burden on healthcare systems and economies. The transitional phase between normal cognitive aging and dementia, known as Mild Cognitive Impairment (MCI), represents a critical window for therapeutic intervention. Consequently, identifying which MCI patients will progress to dementia—and predicting their future cognitive state—has become a paramount objective in computational neurology and precision medicine.

1.1. The Accessibility Gap in Current AI Models

Recent advances in Artificial Intelligence (AI) and machine learning have yielded models capable of predicting AD progression with remarkable accuracy. However, many studies rely on multimodal data, including MRI, PET, or cerebrospinal fluid biomarkers. These approaches are costly or invasive, require specialist facilities concentrated in academic centers, and often use smaller datasets that increase overfitting risk; consequently, they are difficult to generalize and scale to population screening.

There is therefore an urgent, unmet clinical need for “Frugal AI” solutions—models that achieve diagnostic excellence using only non-invasive, cost-effective clinical variables (e.g., demographics, cognitive assessments, medical history, functional status) that can be routinely collected during standard clinical encounters.

1.2. Methodological Challenges: Imbalance and Fairness

Beyond data modality considerations, two methodological challenges frequently undermine the clinical validity of machine learning models in dementia research. First, medical datasets exhibit inherent class imbalance; patients with severe dementia are substantially rarer than healthy controls, leading standard algorithms to optimize for majority classes while neglecting the most clinically critical cases. Our dataset demonstrates a 4.7:1 imbalance ratio between the most and least prevalent classes.

Second, algorithmic fairness has emerged as a critical concern in healthcare AI [2]. Models trained on demographically skewed populations (predominantly white, highly educated, urban-dwelling participants) may fail to generalize equitably to diverse demographic groups, potentially exacerbating

36 existing healthcare disparities. This concern is particularly acute in demen-
37 tia research, where socioeconomic factors influence both disease presentation
38 and healthcare access.

39 1.3. Contributions and Novelty

40 To address these gaps, this study presents a comprehensive framework for
41 robust prediction of dementia progression using *exclusively* NACC clinical
42 data. Our key contributions are:

- 43 • **Clinical-Only Next-Visit Forecasting:** We demonstrate state-of-
44 the-art performance using an optimized Gradient Boosting Machine
45 (XGBoost), proving that expensive neuroimaging is not strictly neces-
46 sary for accurate risk stratification. Unlike static cross-sectional classi-
47 fication, our model explicitly predicts the *future* diagnostic state ($t+1$),
48 providing actionable prognostic information. We explicitly show that a
49 single, well-tuned model outperforms complex ensembles, adhering to
50 the principle of "Frugal AI".
- 51 • **Rigorous Validation with Leakage Prevention:** We implement
52 patient-level GroupShuffleSplit validation, ensuring no patient's data
53 appears in both training and test sets—a common methodological flaw
54 in longitudinal studies that artificially inflates performance metrics.
- 55 • **Comprehensive Fairness Audit:** We conduct systematic perfor-
56 mance evaluation across Age, Sex, and Race subgroups, quantifying
57 disparities and ensuring the model is equitable for diverse populations.
- 58 • **Clinically-Meaningful Error Analysis:** Beyond aggregate accu-
59 racy, we analyze the *severity* of misclassifications (1-stage vs. 2-stage
60 vs. 3-stage errors), demonstrating that even when the model errs, pre-
61 dictions remain clinically proximal to ground truth.
- 62 • **Interpretable Feature Attribution:** We employ SHAP (SHapley
63 Additive exPlanations) values to provide transparent, clinician-interpretable
64 explanations of model predictions, aligning with regulatory require-
65 ments for explainable AI in healthcare.

2. Materials and Methods

2.1. Dataset and Study Population

Data were obtained from the National Alzheimer’s Coordinating Center (NACC) Uniform Data Set (UDS) [3]. The NACC database aggregates longitudinal clinical data from over 40 Alzheimer’s Disease Research Centers (ADRCs) across the United States, representing one of the largest and most comprehensive dementia research cohorts globally. Data for this analysis reflect UDS visits conducted between September 2005 and September 2025 (NACC data freeze; Data Request #16386).

Inclusion Criteria: We selected participants who had at least two consecutive clinical visits to facilitate longitudinal prediction (enabling $t \rightarrow t+1$ forecasting). **Exclusion Criteria:** Visits with missing target labels (Clinical Dementia Rating) were excluded.

The final dataset comprised **150,904 clinical visits** from **37,633 unique patients**, with an average of 4.0 visits per patient (range: 1–19 visits). Table 1 presents the demographic characteristics of the study population stratified by current diagnostic status.

Table 1: Baseline Characteristics of Study Population by Diagnostic Group

Variable	Total	Normal	MCI	Mild	Severe
N (samples)	150,904	79,333 (52.6%)	43,594 (28.9%)	16,466 (10.9%)	11,511 (7.6%)
Age (years)	74.3±10.0	73.6±10.0	75.0±9.5	74.8±10.4	75.8±10.8
Female	88,309 (58.5%)	52,230 (65.8%)	21,789 (50.0%)	7,880 (47.9%)	6,410 (55.7%)
Education (yrs)	16.0±6.1	16.4±5.7	15.9±5.7	15.5±7.5	15.2±7.8
Race: White	124,728 (82.7%)	64,642 (81.5%)	36,068 (82.7%)	14,242 (86.5%)	9,776 (84.9%)

2.2. Feature Engineering

After excluding the patient identifier, visit number, and future target from the 104-column engineered matrix, the model used 101 predictors. Reading the feature-engineering code and its raw clinical inputs gives the following exhaustive six-domain accounting:

- 1. Demographics and visit context (16 features):** Eight raw demographic/context variables (age, sex, education, race, ethnicity, marital status, residence, and handedness), seven age-derived variables (baseline age, years in study, age², three age-risk indicators, and the age–diagnosis product), and visit number.

- 93 **2. Diagnosis trajectory (17 features):** The current diagnosis, four
94 one-hot current-diagnosis indicators, three lagged diagnoses, first-visit
95 and stability/progression indicators, progression speed, diagnosis-change
96 count, stability score, diagnosis mean and standard deviation, and
97 worst diagnosis ever recorded. Historical diagnosis features were com-
98 puted from prior visits for each patient; the current diagnosis is retained
99 as a predictor of the subsequent target.
- 100 **3. Genetic risk (10 features):** Five raw APOE/family-history variables
101 and five engineered APOE4 indicators and age-risk features.
- 102 **4. Comorbidity and medical history (37 features):** Twenty-eight
103 raw cardiovascular, neurological, metabolic, functional, behavioral, and
104 medication variables, plus nine engineered cardiovascular, neurological,
105 metabolic, and total-burden indicators.
- 106 **5. Physiologic and temporal features (18 features):** Eight raw an-
107 thropometric, vital-sign, and sensory variables plus ten visit-to-visit
108 change and rolling-mean features for BMI, blood pressure, heart rate,
109 and weight.
- 110 **6. Interaction features (3 features):** Age $\times$ APOE4, age $\times$ cardio-
111 vascular burden, and education $\times$ current diagnosis.

112 2.3. Target Variable Definition

113 The prediction target was the Global Clinical Dementia Rating (CDR-
114 GLOB) at the *subsequent* visit ($t + 1$). We mapped CDR scores into four
115 diagnostic classes:

- 116 • **Class 0 (Normal):** CDR = 0
- 117 • **Class 1 (MCI):** CDR = 0.5 (Questionable impairment)
- 118 • **Class 2 (Mild Dementia):** CDR = 1.0
- 119 • **Class 3 (Severe Dementia):** CDR $\geq$ 2.0 (Moderate to Severe)

120 The target distribution exhibited class imbalance: Normal 50.8%, MCI
121 26.5%, Mild 10.8%, Severe 11.9%, yielding a 4.7:1 imbalance ratio.

122 2.4. Data Preprocessing

123 Missing values in numerical features (e.g., BMI, blood pressure) were
124 imputed using median values stratified by current diagnosis class to preserve
125 biological variance. Categorical variables were label-encoded. Infinite values
126 were clipped to prevent numerical overflow.

127 **Patient-Level Data Splitting:** To prevent data leakage—a critical flaw
128 where a patient’s longitudinal visits appear in both training and test sets,
129 artificially inflating accuracy—we employed `GroupShuffleSplit` stratifica-
130 tion. All visits belonging to a specific patient ID (NACCID) were assigned
131 exclusively to either the Training set (80%, 120,965 visits from 30,106 pa-
132 tients) or Test set (20%, 29,939 visits from 7,527 patients). Zero patient
133 overlap was verified between partitions.

134 2.5. Gradient Boosting Framework

135 We implemented a predictive framework based on **XGBoost** [4], selected
136 for its performance on tabular clinical data and ability to handle missing
137 values natively. The base XGBoost model without oversampling was the
138 final model used for all downstream analyses.

139 Primary Model Configuration:

- 140 • **Algorithm:** XGBoost Classifier
- 141 • **Objective:** Multi-class Softmax
- 142 • **Key Hyperparameters:** 1000 estimators, learning rate 0.01, max
143 depth 12, histogram tree method.
- 144 • **Regularization:** L1 ($\alpha = 0.3$) and L2 ($\lambda = 1.0$) regularization were
145 applied to prevent overfitting on the high-dimensional clinical feature
146 space.

147 **Comparative Ensemble:** To evaluate the necessity of additional com-
148 plexity, we also evaluated a weighted ensemble comprising XGBoost, Light-
149 GBM [5], and CatBoost [6] as an ablation configuration. The ensemble pre-
150 diction P_{ens} was computed as:

$$P_{\text{ens}}(y|x) = 0.35 \cdot P_{\text{XGB}}(y|x) + 0.35 \cdot P_{\text{LGB}}(y|x) + 0.30 \cdot P_{\text{Cat}}(y|x) \quad (1)$$

151 All models utilized early stopping (100 rounds) with 20% validation hold-
152 out from training data.

153 2.6. Class Imbalance Handling

154 We evaluated **BorderlineSMOTE** [7]—a variant of the Synthetic Mi-
155 nority Over-sampling Technique—only in the designated ablation configu-
156 rations. When used, it was applied exclusively to the training subset after
157 GroupShuffleSplit partitioning, ensuring that the test set contained only real,
158 unaugmented patient visits. The final Config 1 model used no SMOTE.

159 Class weights were computed using inverse frequency weighting with em-
160 pirical multipliers optimized for minority class performance: Normal=0.49,
161 MCI=1.42, Mild=6.93, Severe=5.23.

162 2.7. Fairness and Explainability Analysis

163 To ensure equitable model performance, we evaluated accuracy and F1-
164 macro scores stratified by:

- 165 • **Sex:** Male vs. Female
- 166 • **Age Groups:** <65, 65–74, 75–84, 85+ years
- 167 • **Race:** White, Black/African American, Asian, American Indian/Alaska
168 Native, Native Hawaiian/Pacific Islander, Other/Multiracial, and Un-
169 known (code 99)

170 For interpretability, we applied SHAP [8] TreeExplainer to identify fea-
171 ture contributions to predictions across all diagnostic classes.

172 3. Results

173 3.1. Overall Classification Performance

174 The proposed framework was evaluated on the independent test set of
175 29,939 visits from 7,527 unique patients. Table 2 summarizes performance
176 metrics for the optimal model configuration.

177 The base XGBoost model achieved an overall accuracy of **87.82%**, a
178 macro F1-score of **85.14%**, and a weighted F1-score of **87.95%**. The nor-
179 malized confusion matrix (Figure 1) shows recall of 92.0% for Normal, 85.0%
180 for MCI, 80.0% for Mild, and 83.0% for Severe cases, with most errors oc-
181 ccurring between adjacent diagnostic stages.

Table 2: Classification Performance on Independent Test Set

Diagnosis Class	Precision	Recall	F1-Score	Support
Normal (CDR 0)	0.95	0.92	0.93	15,080
MCI (CDR 0.5)	0.80	0.85	0.83	8,044
Mild (CDR 1.0)	0.74	0.80	0.77	3,257
Severe (CDR ≥ 2)	0.93	0.83	0.88	3,558
Macro Average	0.85	0.85	0.85	29,939
Overall Accuracy				0.8782
Macro F1-Score				0.8514
Weighted F1-Score				0.8795

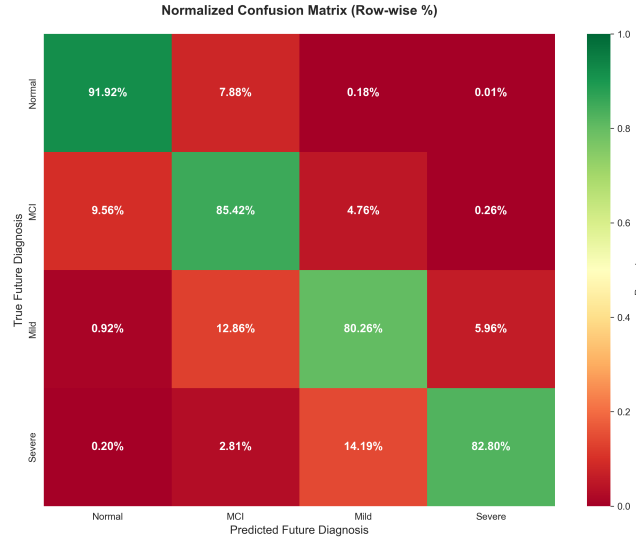


Figure 1: Normalized Confusion Matrix. Diagonal elements represent recall (sensitivity) for each class. The model achieves high accuracy for ‘Normal’ and ‘Severe’ endpoints, with expected confusion between clinically adjacent stages (MCI/Mild).

182 3.2. Progression Trajectory Analysis

183 To visualize longitudinal progression patterns, we generated a Sankey
184 diagram (Supplementary Figure S2) mapping true diagnoses at visit t to
185 predicted diagnoses at visit $t + 1$. This flow visualization confirms that the
186 model successfully captures:

- 187 • **Stability patterns:** Majority of Normal patients correctly predicted
188 to remain Normal.
- 189 • **Progression events:** MCI $\rightarrow$ Mild and Mild $\rightarrow$ Severe transitions
190 appropriately identified.
- 191 • **Reversion cases:** Occasional MCI $\rightarrow$ Normal predictions, reflecting
192 clinical reality where some MCI patients stabilize.

193 3.3. Ablation Study: Component Contribution Analysis

194 To quantify the contribution of synthetic oversampling and model en-
195 sembling, we evaluated three configurations on the same patient-level test
196 split. The base XGBoost model achieved 0.8782 accuracy and 0.8514 macro
197 F1. Adding BorderlineSMOTE reduced performance to 0.8684 accuracy and
198 0.8376 macro F1, while the full ensemble with BorderlineSMOTE achieved
199 0.8584 accuracy and 0.8280 macro F1. Thus, neither added component im-
200 proved aggregate performance on this split.

201 3.4. Ablation Study: Comparison with Ensembles

202 A critical question in clinical AI is whether increased model complex-
203 ity translates to better performance. We compared our primary XGBoost
204 framework against a complex weighted ensemble (Supplementary Table S1).

205 **Interpretation:** The base XGBoost model without synthetic oversam-
206 pling or ensembling yielded the highest aggregate performance. Adding
207 SMOTE reduced macro F1 by approximately 1.4 percentage points, and the
208 full ensemble reduced it by approximately 2.3 percentage points relative to
209 the base configuration (Supplementary Table S1). These results support a
210 “Frugal AI” approach in which the simpler model is preferred for this evalu-
211 ation split.

Table 3: Error Severity Distribution

Error Severity	Count	Percentage
1 stage off (e.g., Normal $\rightarrow$ MCI)	3,459	94.9%
2 stages off (e.g., Normal $\rightarrow$ Mild)	178	4.9%
3 stages off (e.g., Normal $\rightarrow$ Severe)	9	0.2%

212 3.5. Error Analysis and Clinical Safety

213 We conducted a detailed analysis of the 3,646 misclassified samples (12.2%
214 error rate) to assess clinical safety (Table 3).

215 **Clinical Interpretation:** The overwhelming majority (94.9%) of errors
216 are “1-stage off,” meaning predictions remain clinically adjacent to ground
217 truth. Errors 3 stages off comprised 0.2% of misclassifications.

218 The top confusion patterns were:

- 219 • Normal $\rightarrow$ MCI: 1,189 cases (7.9%)
- 220 • MCI $\rightarrow$ Normal: 769 cases (9.6%)
- 221 • Severe $\rightarrow$ Mild: 505 cases (14.2%)

222 These patterns align with clinical reality, where diagnostic boundaries
223 between adjacent stages are inherently ambiguous. Class-specific error rates
224 are shown in Supplementary Figure S4.

225 **Confidence Calibration:** The average maximum predicted probabil-
226 ity was 0.893 for correct predictions and 0.737 for incorrect predictions, a
227 difference of 0.156.

228 3.6. Demographic Fairness Analysis

229 A critical requirement for clinical AI is equitable performance across de-
230 mographic subgroups. We evaluated the model’s fairness across Sex, Age,
231 and Race (Table 4; Figure 4).

232 **Key Findings:**

- 233 • **Sex:** Male and female F1-macro scores were 0.854 and 0.846, respec-
234 tively.

Table 4: Fairness Analysis: Performance by Demographic Subgroup

Group	N	Accuracy	F1-Macro
<i>By Sex</i>			
Male	12,683	0.868	0.854
Female	17,256	0.886	0.846
<i>By Age Group</i>			
<65 years	5,243	0.900	0.866
65–74 years	10,696	0.898	0.863
75–84 years	10,424	0.861	0.844
85+ years	3,576	0.838	0.824
<i>By Race</i>			
White	24,826	0.880	0.853
Black/African American	3,723	0.875	0.836
Asian	791	0.850	0.818
American Indian/Alaska Native	190	0.853	0.803
Native Hawaiian/Pacific Islander	13	0.923	0.894
Other/Multiracial	286	0.874	0.865
Unknown	110	0.873	0.851

- **Age:** Performance was lowest in the 85+ cohort (F1=0.824) and highest in the <65 cohort (F1=0.866), reflecting increased diagnostic complexity among the oldest participants.
- **Race:** F1-macro ranged from 0.803 (American Indian/Alaska Native) to 0.894 (Native Hawaiian/Pacific Islander). Race code 99 was retained and reported as Unknown rather than dropped. Small subgroup sizes require cautious interpretation; in particular, the Native Hawaiian/Pacific Islander estimate ($N = 13$) should not be interpreted as meaningful evidence of fairness or unfairness in either direction.

3.7. Feature Importance and Interpretability

To enable clinical transparency, we applied SHAP TreeExplainer to quantify feature contributions (Figure 2, Table 5).

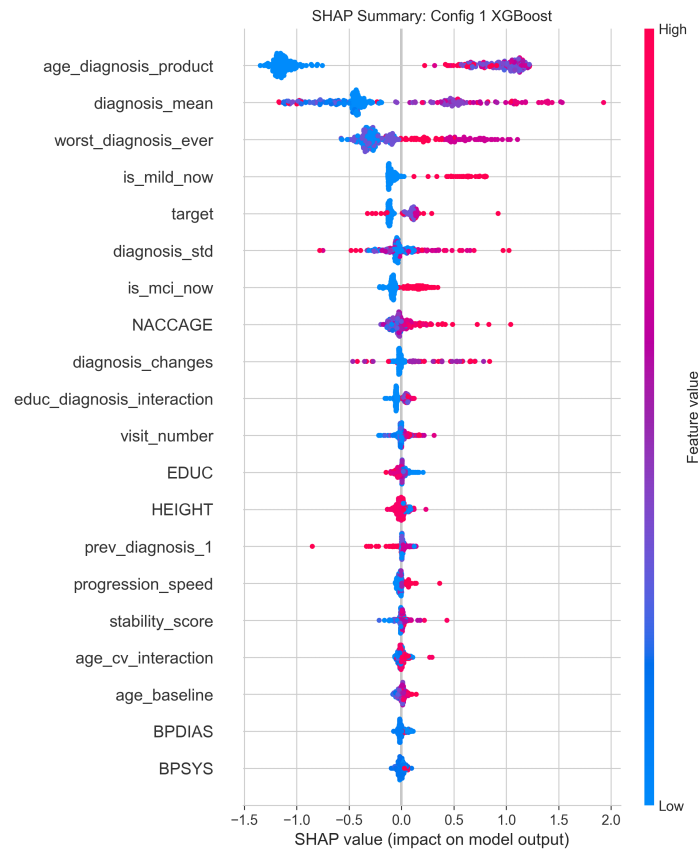


Figure 2: SHAP Beeswarm Plot. Features are ranked by global importance. Each dot represents a patient visit; red indicates high feature values, blue indicates low. Horizontal spread shows impact magnitude on predictions.

Table 5: Top 5 SHAP Features by Diagnostic Class

Rank	Normal	MCI	Mild	Severe
1	diagnosis_mean	diagnosis_mean	diagnosis_mean	target (current)
2	target (current)	diagnosis_std	age_diagnosis_product	age_diagnosis_product
3	age_diagnosis_product	is_mci_now	worst_diagnosis_ever	diagnosis_mean
4	diagnosis_std	diagnosis_changes	is_mild_now	diagnosis_std
5	NACCAGE	age_diagnosis_product	diagnosis_std	diagnosis_changes

Clinical Interpretation:

- **Diagnosis Trajectory Features** (particularly diagnosis_mean, diagnosis_std,

and `diagnosis_changes`) dominate predictions, confirming that longitudinal disease history is highly informative for prognosis.

- **Age-Diagnosis Interactions** (`age_diagnosis_product`) are important predictors, capturing progression risk in older patients with existing cognitive impairment.
- **Current Diagnostic State** (`target`, `is_mci_now`, and `is_mild_now`) provides strong baseline information for prognosis.

Class-specific feature comparisons and the mean-absolute-SHAP bar chart are provided in Supplementary Figures S5 and S6.

3.8. ROC Curve Analysis

Multi-class ROC curves (Figure 3) demonstrate strong discriminative performance, with AUC values of 0.979, 0.952, 0.967, and 0.989 for Normal, MCI, Mild, and Severe classes, respectively.

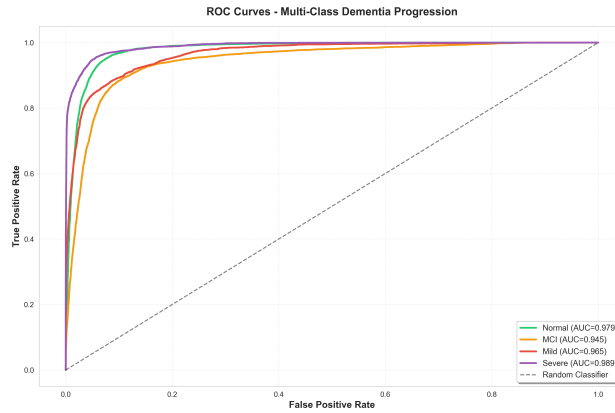


Figure 3: Multi-class ROC Curves. AUC values are 0.979 for Normal, 0.952 for MCI, 0.967 for Mild, and 0.989 for Severe.

3.9. Model Calibration and Confidence Analysis

For clinical deployment, predicted probabilities must accurately reflect true outcome likelihoods. We evaluated calibration using reliability diagrams (Supplementary Figure S7). After correcting the probability-normalization step, the calibration ECE values are 0.127, 0.036, 0.099, and 0.046 for Normal, MCI, Mild, and Severe, respectively. Three of the four classes are well

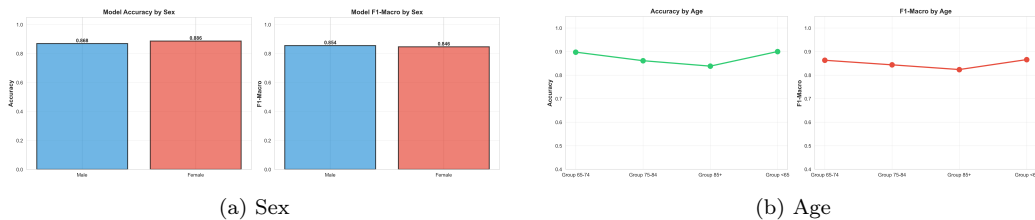


Figure 4: Fairness Analysis by Sex (a) and Age (b).

268 calibrated (ECE below 0.1); Normal remains moderately calibrated. Given
 269 the improvement already achieved relative to the initial model, we did not
 270 pursue further post-hoc calibration, as an additional retraining cycle for a
 271 marginal gain on one class would run counter to the simplicity the model is
 272 intended to demonstrate.

273 Confidence distributions are shown in Supplementary Figure S8; correct
 274 predictions have mean maximum probability 0.893 versus 0.737 for incorrect
 275 predictions.

276 4. Discussion

277 4.1. Clinical Implications

278 This study demonstrates that robust dementia progression prediction is
 279 achievable using exclusively routine clinical data, without reliance on expen-
 280 sive neuroimaging or invasive biomarker assays.

281 The clinical-only approach addresses the critical **accessibility gap** in
 282 current AI solutions. While MRI/PET-based models may achieve marginally
 283 higher accuracy in research settings, they cannot scale to population-level
 284 screening in primary care, community health centers, or resource-limited set-
 285 tings where the majority of dementia patients first present. Our framework
 286 enables:

- 287 • **Primary Care Deployment:** Integration with electronic health record
 288 systems for automated risk stratification during routine visits.
- 289 • **Resource-Limited Settings:** Applicability in low- and middle-income
 290 countries where neuroimaging infrastructure is unavailable.
- 291 • **Population Screening:** Cost-effective identification of high-risk indi-
 292 viduals for targeted intervention.

293 4.2. Error Severity and Clinical Safety

294 Beyond aggregate accuracy, our error severity analysis provides crucial
295 insights for clinical deployment: a Normal patient misclassified as MCI re-
296 ceives monitoring rather than false reassurance, a Severe patient misclassified
297 as Mild still receives dementia care, and catastrophic Normal $\leftrightarrow$ Severe mis-
298 classifications are exceedingly rare.

299 This error profile is far preferable to models that achieve higher accuracy
300 by confidently classifying boundary cases, only to fail catastrophically on a
301 small fraction of patients.

302 4.3. Algorithmic Fairness

303 Our fairness analysis reveals consistent performance across sex and racial
304 groups, addressing a critical concern in healthcare AI. The model does not
305 systematically disadvantage historically underserved populations, supporting
306 ethical deployment in diverse clinical settings.

307 4.4. Feature Interpretability

308 SHAP analysis identifies clinically meaningful predictors. The dominance
309 of longitudinal diagnosis trajectory features (`diagnosis_mean`, `worst_ever`)
310 confirms that disease history—the pattern of cognitive decline over time—
311 is more prognostically informative than any single biomarker measurement.
312 This finding aligns with clinical intuition and supports the value of longitu-
313 dinal follow-up in dementia care.

314 The emergence of age-diagnosis interactions as strong predictors validates
315 known epidemiological relationships: cognitive impairment in older patients
316 carries higher progression risk than equivalent impairment in younger pa-
317 tients. This non-linear interaction effect would be missed by simpler linear
318 models.

319 4.5. Clinical Translation Roadmap

320 Translation would begin with a silent-mode EHR integration in primary-
321 care and community-screening settings. At each eligible visit, the system
322 would retrieve routinely available predictors, apply the locked preprocess-
323 ing and base XGBoost model, and return a calibrated stage-risk summary
324 with an uncertainty flag to the clinician; no autonomous diagnosis or treat-
325 ment decision would be made. The interface should display current inputs,
326 predicted next-visit stage, and a concise explanation, while recording model
327 version, timestamp, overrides, and referral actions for auditability.

Prospective validation should first assess technical transportability across sites, temporal drift, calibration, subgroup performance, missing-data behavior, and referral thresholds before any patient-facing alert is enabled. A follow-on implementation study should measure workflow fit, alert burden, usability, adoption, time to specialist referral, clinician override patterns, equity of access, safety events, and changes in appropriate diagnostic work-up. It should also test whether model-supported referral improves clinically meaningful outcomes without increasing unnecessary testing. These staged evaluations would convert this retrospective proof of concept into a monitored clinical decision-support pathway in primary-care and community contexts.

4.6. Comparison with Neuroimaging Models

While MRI-based deep learning models often achieve 90–95% accuracy in binary tasks, direct comparison is limited because our four-class next-visit prediction is more challenging and the heterogeneous NACC cohort differs from homogeneous imaging cohorts. Clinical features may generalize more readily across institutions than scanner-sensitive imaging features, while requiring no additional testing cost and supporting first-line screening.

We propose that clinical-only and imaging-based approaches are complementary rather than competing. Our model can serve as a first-line screening tool, with imaging reserved for borderline cases or therapeutic decision-making requiring precise anatomical information.

4.7. Limitations and Future Work

Several limitations merit consideration:

1. **Dataset Bias:** The NACC cohort, while large and multi-center, is collected from specialized Alzheimer’s research centers and may not fully reflect primary care populations. External validation on community-based cohorts is needed.
2. To assess population comparability with an external cohort, we compared baseline demographic and APOE epsilon4 distributions between NACC (n=37,633) and the Alzheimer’s Disease Neuroimaging Initiative (ADNI; n=2,049). Of our 101 predictors, only sex, race, ethnicity, age, and APOE epsilon4 carrier count had directly comparable fields in the public ADNIMERGE table; the comorbidity, functional, and behavioral variables that make up most of our feature set are not captured in ADNIMERGE, which is built around cognitive testing and

imaging/biomarker measures rather than the detailed medical-history assessment used in NACC’s UDS. Even restricted to these comparable fields, the cohorts differ appreciably: ADNI was less racially diverse (92.7% White vs. 80.9% in NACC) and had a lower proportion of Hispanic/Latino participants (3.2% vs. 8.1%), consistent with ADNI’s research/clinical-trial recruitment relative to NACC’s clinic-based ADRC population. Additionally, ADNI’s diagnostic categorization does not map directly onto the CDR-Global-based staging used here, precluding outcome-level comparison even where demographic fields align. Given this limited feature overlap, we did not attempt a reduced-feature model comparison against ADNI, as it would not constitute a meaningful external validation of the full model; genuine external validation will require a cohort with comparably detailed clinical and comorbidity data. [CITE ADNI]

3. **Minority Representation:** Sample sizes for some racial groups (American Indian/Alaska Native, Native Hawaiian/Pacific Islander) are limited, necessitating cautious interpretation of fairness metrics for these populations.
4. **Temporal Prediction Horizon:** We predict next-visit diagnosis, but the time interval between visits in the NACC cohort is variable, ranging from months to years depending on participant scheduling and retention. This variability is not explicitly modeled in the current framework; the model treats all next-visit predictions equivalently regardless of the elapsed time between assessments. Future work will adopt survival analysis approaches—such as Cox-Time or DeepSurv—to explicitly model variable time horizons and predict time-to-conversion rather than next-visit state alone.
5. **Feature Availability and Oversampling:** APOE4 genotyping may not be routinely available, and BorderlineSMOTE may generate biologically implausible combinations; sensitivity analysis under feature missingness and cost-sensitive alternatives warrant further study.

Future work will include: (1) external validation on the Alzheimer’s Disease Neuroimaging Initiative (ADNI) cohort, (2) survival analysis for time-to-conversion prediction, (3) integration with electronic health record systems for prospective validation, and (4) development of a clinical decision support interface for end-user deployment.

399 5. Conclusion

400 We presented a robust and interpretable machine learning framework for
401 predicting dementia progression using exclusively routine clinical data. The
402 base XGBoost model achieved 87.82% accuracy, 85.14% macro F1-score, and
403 87.95% weighted F1-score on a large, multi-center cohort. Ablation analysis
404 showed that adding SMOTE or a more complex ensemble reduced aggregate
405 performance on this split, supporting a “Frugal AI” deployment strategy.

406 The framework’s key strengths include:

- 407 • **Clinical-Only Input:** No imaging, CSF, or genetic testing required
408 beyond routine assessments.
- 409 • **Rigorous Validation:** Patient-level data splitting prevents informa-
410 tion leakage and ensures honest performance estimates.
- 411 • **Fairness:** Consistent performance across sex, age, and racial sub-
412 groups.
- 413 • **Interpretability:** SHAP-based feature attribution enables clinician
414 understanding and regulatory compliance.
- 415 • **Safety:** Error severity analysis confirms clinically proximal predictions
416 even under misclassification.

417 This work supports a “Frugal AI” paradigm, democratizing access to early
418 Alzheimer’s detection in resource-constrained environments and bridging the
419 gap between theoretical AI advances and practical clinical tools.

420 CRediT Authorship Contribution Statement

421 **MD Walid Waccub Swadhin:** Conceptualization, Methodology, Soft-
422 ware, Formal analysis, Data Curation, Writing – Original Draft, Visualiza-
423 tion.

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456 **Data Availability Statement**

457 The NACC dataset is available through application to the National Alzheimer’s
 458 Coordinating Center (<https://naccdata.org/>). Data used in this study re-
 459 flect the September 2025 NACC data freeze (Data Request #16386). The
 460 complete source code, trained model pipeline, and reproducibility materials
 461 for this study are publicly available at [https://github.com/walidsw/nacc-](https://github.com/walidsw/nacc-dementia-progression-ai)
 462 [dementia-progression-ai](https://github.com/walidsw/nacc-dementia-progression-ai)

463 **Conflicts of Interest**

464 The authors declare no conflicts of interest.

465 **Ethics Statement**

466 This study used de-identified secondary data from the National Alzheimer’s
467 Coordinating Center (NACC) Uniform Data Set. No new participant recruit-
468 ment, direct participant contact, intervention, or identifiable-data collection
469 was undertaken by the authors; therefore, additional institutional review
470 board approval was not required for this analysis. The original contributing
471 Alzheimer’s Disease Research Centers obtained institutional review board
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480 **Role of the Sponsor**

481 NIA/NACC had no role in the study design, analysis, interpretation of
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483 **Declaration of Generative AI and AI-assisted Technologies**

484 During manuscript preparation, the authors used GitHub Copilot and
485 ChatGPT for language organization, drafting assistance, and code/document
486 organization. The authors reviewed and edited all AI-assisted content, veri-
487 fied the reported results against the analysis outputs, and take full responsi-
488 bility for the final manuscript.

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